

PERFECTOID FIELDS and THE TRANSFER THEOREM

Function field analogy

$$\mathbb{Q}_p \quad \mathbb{F}_p((t))$$

two approach to do that : $\begin{cases} p \rightarrow \infty \\ e \rightarrow \infty \end{cases}$

if $p \rightarrow \infty$ and consider an ultrafilter on primes

$$\prod_{p \rightarrow u} \mathbb{Q}_p / \mathcal{U} \quad \stackrel{\text{AKE}}{=} \quad \prod_{p \rightarrow u} \mathbb{F}_p((t)) / \mathcal{U}$$

henselian valued fields
of char $(0,0)$

ARTIN'S CONJECTURE

C_i -fields (homogeneous polynomials of $n > \sum d_i$ have always solutions)
 $\downarrow$ degrees
 $\uparrow$ # of variables

$$\mathbb{F}_p((t)) \text{ is } C_2$$

$\rightarrow$ is $\mathbb{Q}_p$ C_2 ? No

if $e \rightarrow \infty$ (ramification) $e = \#(0, \nu(p)]$

$$(0, \nu(p)] \subseteq \nu K$$

DEFINITION (perfectoid fields) $\rightarrow$ of residue char $p > 0$

A valued field (K, ν) is perfectoid

(i) $\forall K$ rank 1 ($\leq \mathbb{R}$) and (K, ν) complete

(ii) $\forall K \not\cong \mathbb{Z}$

(iii) $(\mathcal{O}_K/\mathfrak{p})$ semiperfect, i.e. Frob $(\mathcal{O}_K/\mathfrak{p}) \rightarrow (\mathcal{O}_K/\mathfrak{p})$

(perfect if Frob is bijective)

$$x \mapsto x^p$$

the residue fields stay the same

is surjective

$[0, \nu(p)]$ is p -divisible

it contains something not trivial by (ii) $\Rightarrow$ and everything is p -divisible

EXAMPLES

- compatible system of p -th roots and then it is an ∞ extension. some complete it $\mathbb{Q}_p(p^{1/p^\infty})$
- $\mathbb{F}_p((t^{1/p^\infty}))$
- $\mathbb{Q}_p(p^{1/p^\infty})$ 1 compatible system of p -th ^{primitive} root of unity
- ~~$\mathbb{Q}_p(p^{1/p^\infty})$~~

80's fountain - Wintenberger

$$G_{\mathbb{Q}_p(p^{1/p^\infty})} \cong G_{\mathbb{F}_p((t^{1/p^\infty}))}$$

Kholue: perfectoid fields

the idea is that perfectoid fields are generalisation of where this $\cong$ still holds

methods: tilting

idea:

perfectoid field $\xrightarrow{\text{tilt}}$ perfectoid field
char $(0, p)$ char (p, p)

DEFINITION (tilt) $K \mapsto K^b$

let (K, r) be perfectoid

$$K^b := \varprojlim_{x \mapsto x^p} K$$

inverse limit in the monoid category with multiplication

$$\cdots \leftarrow K \xleftarrow{x^p \mapsto x} K \xleftarrow{x^p \mapsto x} K \xleftarrow{x^p \mapsto x} \cdots$$

$$\mathbb{O}_N^b := \varprojlim_{x \mapsto x^p} \mathbb{O}_N = \varprojlim_{x \mapsto x^p} \varprojlim_{n \in \mathbb{N}} \mathbb{O}_N / (p^n)$$

$$\mathbb{O}_N \text{ p-adically complete } \mathbb{O}_N = \varprojlim_{n \in \mathbb{N}} \mathbb{O}_N / (p^n)$$

Fubini

$$= \varprojlim_{n \in \mathbb{N}} \varprojlim_{x \mapsto x^p} \mathcal{O}_v / (p^n) = \varprojlim_{\substack{n \in \mathbb{N} \\ x \mapsto x^p}} \mathcal{O}_v / (p^n) \quad \leftarrow \text{also as rings}$$

$\leadsto$ ring structure on $\mathcal{O}_v^b \rightarrow (\text{char} = p)$

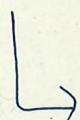
$$K^b := \text{Frac}(\mathcal{O}_v^b) \supseteq \mathcal{O}_v^b$$

the multiplication coincides with the monoid one of the previous inverse limit

$$\# : \mathcal{O}_v^b \rightarrow \mathcal{O}_v$$

$$(x_n)_{n \in \mathbb{N}} \mapsto x_0$$

$$\in \varprojlim_{x \mapsto x^p} \mathcal{O}_v$$



$$N^b(x) := N(\#(x)), \quad \mathcal{O}_v^b = \mathcal{O}_{v^b}$$

THEOREM (Fontaine Wintenberger)

(K, v) perfectoid

$$\Rightarrow G_K \cong G_{K^b}$$

(@p it is not convoluted enough in fact $\mathcal{O}_p^b = \mathbb{F}_p$)

~~perfectoid~~

TRANSFER THEOREM

(K, v) henselian valued field of char $(0, p)$

such that $\cdot G_K$ is small

$$\cdot \text{Conv}(v_p) = v_K \text{ sur}$$

$\cdot v_K$ p -divisible

Then $\exists F$ with $\text{char}(F) = p$ & $G_F \cong G_K$

Lemma 4.9

Let (K, v) valued field $\text{char}(O, p) \neq 2$.

(a) $\dim_{\mathbb{F}_p} K^x / (K^x)^p \propto \infty$

(ii) v_K p -divisible rank 1

(iii) K_x perfect

$\Rightarrow G_v / (p)$ semiperfect, $(\hat{K}, \hat{r})$ perfectoid

proof

from (i) $|K^x / (K^x)^p| < \infty \Rightarrow$ choose representatives

$$v \in K \text{ p-divisible} \quad a_1(k^x)^p, \dots, a_n(k^x)^p$$

$\sqrt{k}$ p-durable

$\Rightarrow$ I can make multiplications
to that the representative
have valuation 0

WLOG

wlog $\Rightarrow a_1, \dots, a_n \in \mathcal{O}_v^x$ and a_1

by perfection of $K\pi \leadsto a_2, \dots, a_n \in 1 + m_V$

Let $\gamma = \min_{i \in \{1, \dots, n\}} v(a_i^0 - 1) > 0$

$$\text{rk rank } 1 \Rightarrow m\gamma \geq r(p) \text{ for some } m \in \mathbb{N}$$
$$\forall x \in G_v$$

$$x = y_1^p (1 + z_1) \quad \text{where } z_1 \in \{a_1^{-1}, \dots, a_n^{-1}\}$$

Inductively $z_i^0 = y_{i+1}^P (1 + z_{i+1}^0)$ for $i = 1, \dots, m$

$y_1, \dots, y_m \in Q$

$$x = y_1^p (1 + y_2^p (1 + y_3^p (\dots))) = y_1^p + y_1^p y_2^p + \dots + y_1^p y_2^p \dots y_m^p z_m$$

$$v(y_1^p y_2^p \dots y_m^p z_m) = v(x) + v(z_1) + \dots + v(z_m) \geq m\gamma \geq v p$$

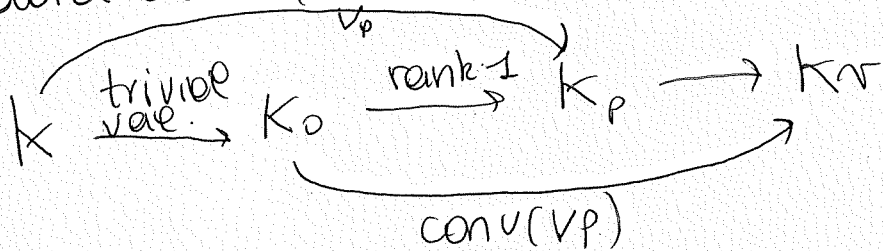
$\Rightarrow$ since we
are in $\mathcal{O}_V/(p)$ $y_1 + y_1 y_2 + \dots + y_1 y_2 \dots y_m$ works

□

proof of the transfer theorem

if G_K is small $\xRightarrow{3.19} \dim_{\mathbb{F}_p} K^\times / K^{\times p} < \infty$

standard decomposition



consider ν_p : rank 1, $\nu_p K$ p -divisible

POPIL LEMMA: K_p is perfect

$\xRightarrow{4.9} (\hat{K}, \hat{\nu}_p)$ perfectoid

$$G_K \cong_{\text{Krauer}} G_{\hat{K}} \cong_{\text{FW}} G_{\hat{K}^b}$$

take $F = \hat{K}^b$

□

COROLLARY 4.28

let (K, v) henselian valued field $\text{char}(O, p)$

Assume G_K small

v_K regular above $v(p)$ & p -divisible

$\Rightarrow \exists F$ with $\text{char} F = p$

$$G_F \cong G_K$$

i.e. $v_K / \text{con}(v_p)$

divisible
stable under
first order

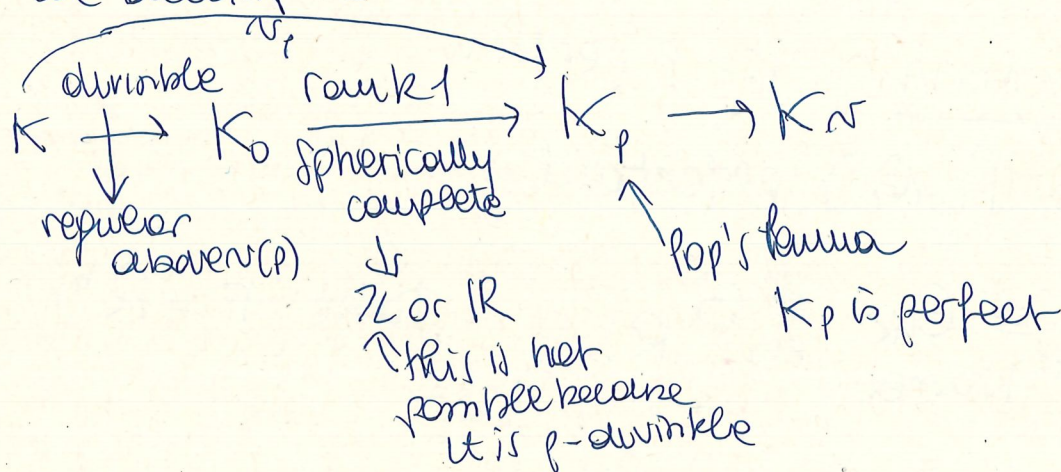
proof

Klingen 17A : Small Galois groups
are preserved with elementary equivalence

WMA K is λ_1 -saturated

saturation composition methods

Now we decompose



(K, v_p) absolutely unramified

i.e. $K \cap v_p$ perfect, ~~regular~~

(K, v_p) henselian, $I_K = 1$

$$G_K = D_K \cong G_{K \cap v_p} / I_K \cong G_{K \cap v_p}$$

↑
henselian

you can take K_p

□